

Strabismus

HISTORY:

Presenting Complaints:

- Ocular Deviation (eso-, exo-, etc.)
- Diplopia

Details

Ocular deviation

- Onset: Congenital or Acquired - precipitating event)
- Duration
- Constant or intermittent
- Diurnal variation
- Precipitating event
 - Trauma
 - Ocular surgery (cataract surgery, scleral bucking)
 - Fever, convulsions, focal neurological deficits
 - Debilitating illness
- Associated eye strain, headache and watering
- Specific to Intermittent Exodeviation
 - Percentage of time strabismic: more than 50%
 - Present in attentive state
 - Lasting through a blink

Very often a pre-existing ocular deviation might get noticed after a **trivial traumatic event**, when the eye was “looked at” in detail.

Diplopia

- Monocular or Binocular
- Horizontal or Vertical
- Maximum in which position of gaze
- More for distance or near
- Does the condition interfere with school, work, play, etc.?
- Aggravating or relieving factors
- Associated vertigo or past pointing
- Abnormal head posture (could occur in absence of subjective diplopia)

Past History

- Spectacle use
 - Date of last prescription
 - Power of present glasses
 - Use of occlusion therapy previously
- Migraine
- Trauma
- Ocular surgery
- Fever, convulsions, focal neurological deficits
- Debilitating illness

Medical History

- Diabetes

Family history

Antenatal, Birth and Developmental history (in congenital strabismus)

- Maternal illness, medication, fever with rash during pregnancy
- Route of birth, forceps assisted delivery, low birthweight, perinatal anoxia
- Developmental milestones (motor and intellectual – schooling)

EXAMINATION:

General Physical Examination:

- Shape of head, facial asymmetry, dysmorphic features (cyclovertical paralyses and craniostenoses)
- Other deformities like extra digits

Systemic Examination:

- Pay special attention to CNS examination
- Details of some cranial nerves will be presented with ocular examination

OCULAR EXAMINATION:

Head posture:

- Head tilt
- Chin elevation
- Face turn

Ocular Posture:

- General Ocular Inspection (Exo- or enophthalmos)
- Hirschberg's Test
- Cover/Uncover test
 - Tropia or Phoria
 - Concomitant or Incomitant (Primary and Secondary deviation)
 - Nystagmus: type, jerky or pendular, direction, null point
 - Dissociated Vertical Deviation

For Concomitant Squint

- Alternate Cover Test
 - Free alternation
 - Dominant Eye (preference to fix with which eye)
- Krimsky's Test
 - Objective and subjective
- Prism Bar Cover Test
 - In 9 positions of gaze for distance, plus for near
 - A/V pattern
 - AC/A ratio (approximate)
 - Watch for variable deviations
- AC/A ratio
 - Heterophoria method
 - Lens gradient method
- Measurement of near point of accommodation and near point of convergence
 - RAF rule or equivalent
 - Base-In and Base-Out Fusional Reserve
- Detect cyclodeviation
 - Double Maddox

For Incomitant Squint

- Diplopia charting
- Hess chart or Lees screen
- Park-Beilschowsky's 3-step test for vertical deviation
- Detect cyclodeviation (Double Maddox)

Sensory Examination: retinal correspondence, suppression, stereopsis

- Bagolini's Striated Glasses
- Worth Four Dot Test
- Titmus fly or TNO test

Ocular Examination

- Visual Acuity (Best corrected for distance and near, Colour, Confrontation)
- Lids
 - Palpebral fissures (equal, ptosis)
 - Epicanthal folds
- Anterior Segment Examination (aniridia, coloboma, heterochromia, cataract)
- Pupil: direct and consensual
- Posterior Segment
 - Remember Bruckner's Test
 - Fundus Torsion
 - Eccentric Fixation (Foveal, Parafoveal, Perifoveal)
- Extraocular Movement
 - Duction
 - Versions
 - Look for superior oblique and inferior oblique over action: grading
 - Convergence
 - Forced Duction Test, Forced Generation Test
 - Differential Tonometry
 - Saccadic Velocity
- Refraction

Grading Oblique Over-Action:

1. 15° deviation from horizontal
2. 30° deviation
3. 60° deviation
4. 90° deviation